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RLAX: A Reinforcement Learning-Based Framework for Explaining Anomalous Data

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Abstract

Singling out the most informative features able to discriminate between normal and anomalous data is a formidable problem dealing with the enormous search space represented by the power set of the whole feature set. These features indeed encode the explanation of the deviating behavior associated with selected observations and represent the hidden knowledge of interest for the analyst in many applicative contexts, namely cybersecurity, bioinformatics, and green by AI, to cite a few. In this work, we present RLAX, a reinforcement learning-based algorithm detecting explanations from collections of high-dimensional data sets. The algorithm receives in input the unlabelled data collection of normal instances, a small set of labelled inlier examples, and a small set of anomalies to be explained. Separation between normal samples and anomalies is measured by means of a distance-based outlier score. The designed learning strategy thus induces a ranking among features that allows isolating those maximizing separation accuracy. Beyond an add-only formulation, we investigate a subspace refinement strategy that augments the action space with *remove* and *stop* actions, enabling non-monotone refinement and adaptive early termination. Experimental results highlight that the size of the explanation returned consists of a small fraction of the whole feature set, and that the associated accuracy and subspace margin are of remarkable quality. Additional analyses of the refinement strategy further clarify its impact on exploration efficiency and explanation quality.

Keywords Explainable AI · Anomaly explanation · Reinforcement learning · Feature selection

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1 Introduction

Anomaly detection identifies observations that deviate from normal behavior, but in many applications detecting an anomaly is not sufficient: analysts also need to understand which characteristics are responsible for the deviation. This motivates the *Anomaly Explanation* problem addressed in this work. Given a population of normal data together with a collection of anomalous examples, our goal is to identify an intelligible explanation of why the anomalies deviate from the inlier population.

We represent explanations as compact feature subspaces in which the anomalous examples deviate most from normal data. Such subspaces identify the features characterizing the abnormal behavior and may reveal useful knowledge to the analyst in several application domains, including Cybersecurity, Bioinformatics, and Green by AI.

Identifying the most informative features is challenging because the search space corresponds to the power set of the whole feature set. To tackle this problem, we propose RLAX, for *Reinforcement Learning-based Anomaly eXplanation*, which formulates anomaly explanation as a Markov Decision Process. Given a measure of separation between normal and anomalous data, we define an *Explanation Gain* that trades off separation accuracy and relative subspace size, and a *Subspace Margin* measuring the separation between the densities associated with inliers and outliers.

In its basic formulation, RLAX explores the feature-subset space by progressively adding features and receives as reward the corresponding variation of Subspace Margin. The policy is learned through Proximal Policy Optimization (PPO), whose ability to handle large action spaces is particularly suitable when features correspond to available actions. The outcome of the learning phase is then exploited to rank features and derive candidate anomaly explanations.

A preliminary add-only formulation of RLAX was introduced in our previous conference work (Amirato et al., 2025). The present journal version extends it with a *Subspace Refinement Strategy*, which augments the action space with Remove and Stop in addition to Add. This enables non-monotonic exploration, allowing the agent to revise previous selections and explicitly terminate the search. The journal version also provides a broader empirical assessment of the resulting framework and of the impact of the refinement strategy.

To the best of our knowledge, RLAX is the first approach exploiting Reinforcement Learning to address Anomaly Explanation as a feature selection problem. Experimental results show that RLAX identifies compact explanations while retaining strong discriminative quality, also in comparison with existing subspace-selection methods for anomaly detection.

The rest of the work is organized as follows. Section 2 describes related work, Sect. 3 formalizes the anomaly explanation problem and describes the RLAX algorithm, Sect. 4 presents the experimental evaluation, and Sect. 5 concludes the work.

2 Related Works

Anomaly analysis involves two complementary tasks: *anomaly detection* (Chandola et al., 2009) and *anomaly explanation* (Panjei et al., 2022; Mokoena et al., 2022; Li et al., 2023; Angiulli et al., 2021a, 2021b, 2022a, 2022b, 2023), where the latter aims to determine the

causes that make an instance anomalous. If an anomaly is detected by an algorithm, Explainable Anomaly Detection techniques aim at making the detector interpretable; conversely, if an anomaly is identified by an expert, they aim at explaining its abnormality by extracting hidden knowledge from data. Our work deals with this second family, focusing on feature-based explanations represented by subspaces. Feature selection using reinforcement learning has also been investigated for classification (Gaudel & Sebag, 2010; Hazrati Fard et al., 2013; Sali et al., 2021), but not specifically for anomaly explanation.

The reader is referred to Li et al. (2023) for a survey on explainable anomaly detection. Some approaches do not select interpretable feature subsets, but instead project data onto low-dimensional spaces. For example, SUOD (Zhao et al., 2020) employs random subspace projections to build outlier ensembles, while Sathe and Aggarwal (2016) combines random projection and sketching to estimate local density. Since these approaches map data into a feature space different from the original one, their projections cannot be directly interpreted as feature-based explanations.

Among methods searching directly in the input feature space, the seminal work Aggarwal and Yu (2001) identifies low-density subspaces through genetic algorithms, while Vinh et al. (2016) introduces the Z-score and Isolation Path score (iPath) to address dimensionality bias. However, these approaches do not exploit example anomalies.

A related approach is proposed by Fassetti and Angiulli (2010), which formulates the problem as subspace selection guided by outlier and inlier examples and uses a genetic algorithm to maximize their separation. Other subspace-based approaches include Riahi-Madvar et al. (2022), which combines linear programming and heuristic search to identify sparse local subspaces, and Cheng et al. (2020), which employs an ensemble and self-paced learning to identify relevant attributes separating outliers from inliers.

HiCS (Keller et al., 2012) identifies high-contrast subspaces in which local distributions significantly differ from the global distribution. More recently, Cribeiro-Ramallo et al. (2025) introduced V-GAN, which formulates the Multiple Views effect as a stochastic subspace-selection problem and addresses it through a generative model. SOD (Kriegel et al., 2009) instead identifies a relevant subspace for each data point based on its local neighborhood.

Other anomaly-explanation methods target specific data types, including networks (Xu et al., 2021), sequence data (Chandola et al., 2012; Wang et al., 2020), and categorical data. For the latter, Angiulli et al. (2009) identifies subsets of attributes accounting for anomalous behavior, while CUFS (Pang et al., 2016) performs unsupervised feature selection by modeling dependencies among categorical features.

Different notions of explanation have also been considered. For example, M^2OE (Angiulli et al., 2024) explains anomalous instances through transformations, identifying minimal changes that turn an outlier into a normal instance. Related approaches can also explain groups of outliers (Macha & Akoglu, 2018).

Explainable anomaly detection has also been investigated for non-tabular data, particularly images, where explanations are commonly represented as heatmaps localizing anomalous regions. For example, AE-XAD (Angiulli et al., 2025) employs a semi-supervised Autoencoder to jointly detect anomalies and provide heatmap-based explanations through reconstruction errors. Other representative approaches include PatchCore (Roth et al., 2022), DRAEM (Zavrtanik et al., 2021), and FCDD (Liznerski et al., 2021), which provide different forms of spatial anomaly localization. Differently from these image-oriented

approaches, our work focuses on feature-based explanations represented as interpretable subspaces of the original feature space.

3 Problem Statement and Method

In this section we introduce RLAX algorithm, for Reinforcement Learning-based Anomaly eXplanation. The rest of the section is organized as follows: Sect. 3.1 provides the definition of Anomaly Explanation, Sect. 3.2 introduces the notion of Subspace Margin, Sect. 3.3 details the learning procedure, and Sect. 3.4 describes how the solutions are generated based on the output of the previous phase.

3.1 Anomaly Explanation

A *feature* f is an identifier with an associated domain $\mathbb{D}(f)$. The overall set of features is denoted by $F = \{f_1, \ldots, f_d\}$ and the associated *domain* $\mathbb{D}(F)$ is the Cartesian product $\mathbb{D}(F) = \mathbb{D}(f_1) \times \ldots \times \mathbb{D}(f_d)$. A *subspace* S is a subset of F.

A *dataset* X on the set of features F is a set of elements x coming from $\mathbb{D}(F)$. Given a subspace S of F and an instance x of X, by x^S we denote the projection of x onto the subspace S, while by X^S we denote the set $\{x^S \mid x \in X\}$.

Assume a set E_{pos} of *outlier examples* (or *positive examples*) is given. We are interested in finding an *explanation* for the anomalous behavior of the instances in E_{pos}.

In this work, we encode explanations under the form of those minimal subspaces where the outlier examples deviate mostly from the dataset population. Indeed, such an explanation makes intelligible the features characterizing the abnormality at hand.

Let sep be a measure of the separation quality between outlier and normal examples, that can be the subspace *margin* defined in the sequel of this section or any other different measure for evaluating classification accuracy, like the Area Under the ROC Curve (AUC) which is suitable for the unbalanced classes scenario typical of the anomaly detection task, to be employed as a measure of success in place of the subspace margin for a fair comparison with competitors aiming at optimizing other measures.

An *Anomaly Explanation* $\mathcal{E}$ of E_{pos} with respect to X is the subspace maximizing the value of sep over the dataset, that is to say the one exhibiting better separation between normal and abnormal samples. More in general, we consider the *Explanation Gain* measure, also called simply *Gain*, defined as

$$Gain_\alpha(S) = sep(S) - \alpha \frac{|S|}{|F|}$$

representing a trade-off between the separation accuracy guaranteed by the feature set S and its relative size $|S|/|F|$. Intuitively, putting one additional feature into the solution set should be compensated by a sufficiently large increase in the separation accuracy, increase which is precisely expressed by the quantity $\alpha/|F|$. Ideally $\alpha \leq 1$ and, if not otherwise stated, in the following we use the value $\alpha = 0.5$. We note that for $\alpha = 0$ the above measure is based only on the sep value. Thus, in the following we say that the best feature set is the one maximizing the $Gain$, and refer to the sep measure also as $Gain_0$.

3.2 Subspace Margin

Given an instance x of X, the *anomaly score* $as(x)$ of x is defined as (Angiulli & Pizzuti, 2002):

$$as(x) = \frac{1}{K} \sum_{i=1}^{K} \text{dist}(x, nn_i(x)),$$

where $nn_i(x)$ denotes the i-th nearest neighbor of x in X according to some pre-defined distance measure. Thus, the anomaly score is given by the sum of the distances between x and its K nearest neighbors in the dataset. Its value provides an estimate of the dataset density in the neighborhood of the instance x. Intuitively, the higher the anomaly score associated with an instance, the more likely it is that the instance was generated by a distribution different from that of X. Such instances are called *outliers* or *anomalies*.

In the following, we assume a second set of examples is given, that is the set E_{neg} consisting of *inlier examples* (or *negative examples*), possibly singled out from the dataset X.

In order to measure the relevance of the subspace S as an explanation, next the concept of subspace margin is introduced. The *subspace margin* of the space S with respect to the set of positive examples E_{pos} and the set of negative examples E_{neg} is

$$margin(S) = \frac{as(E_{pos}^S) - as(E_{neg}^S)}{\sqrt{|S|}},$$

where, with a little abuse of notation, the anomaly score $as(E)$ of a set of examples E is defined as the mean of the outlier scores associated with the elements of E. The subspace margin represents the difference between the anomaly scores of the positive and negative examples. Intuitively, the larger the subspace margin, the larger the separation between the densities associated with the two above mentioned set of examples. The denominator acts as a normalizing factor needed to compare distances coming from subspaces having different sizes.

Thus negative examples have a twofold usefulness: first of all they help to reduce the computational burden associated with estimating separation between normal and abnormal data densities and, moreover, they can be optionally pointed out by a domain expert in order to guide the search to focus on some characteristics of normal data of interest for the analysts. Finally, we note that if a set of negative examples is not available, a randomly selected subsample of the normal data X can always be reliably employed.

3.3 Learning Procedure

The RLAX algorithm, for Reinforcement Learning-based Anomaly eXplanation, formalizes the problem of detecting the anomaly explanation for a given set E_{pos} of positive examples as a reinforcement learning task.

In our formalization the notion of *state* S coincides with that of subspace. Specifically, we map the above problem into a Markovian Decision Process $\langle \mathcal{S}, A, T, R, \gamma \rangle$ where $\mathcal{S}$ is the set

of states, that corresponds to the power set of the feature set F, and A is the set of actions, where an action a corresponds to adding a single feature to the current state.

We denote each action also as the feature that must be added, thus doing the action a in the state s produces the next state $s' = s \cup \{a\}$. Hence, the environment is deterministic and the probability $T(s, a, s')$ of transition from s to s' after doing a is always 1 for $s' = s \cup \{a\}$ and 0 otherwise. Clearly, in each state s, only the subset $A \setminus s$ of actions of A is available.

As for the *reward* R, we use the variation of the subspace margin between the increased subspace and the original subspace, that is

$$R(s, a, s \cup \{a\}) = margin(s \cup \{a\}) - margin(s).$$

We note that using the subspace margin during learning has two advantages. First of all, from the computational point of view it *can be more efficiently computed* with respect to measures, like the AUC, that require knowing the ranking of the anomaly scores associated with all the data, thus to compute the anomaly score of all examples and all dataset instances. Indeed, since the above scores have to be recomputed for each explored subspace, in the presence of large datasets the required temporal cost may become prohibitive. The margin instead can be more conveniently estimated by means of sampling procedures, like the one employed here exploiting positive and negative examples as representative samples of the abnormal and normal data populations respectively. Second, since the margin is a density estimation procedure, making decisions based on it enables the algorithm to be *agnostic to different quality measures* that the user could adopt to evaluate the returned explanation or that can be used to compare the quality of our explanations with that of alternative methods.

Each episode starts with the state $s_0 = \emptyset$, associated with the empty set of features, and proceeds by adding a selected feature a until a stopping criterion is met. The criterion for stopping the current episode is the following: either (1) a maximum number t_{max} of steps is reached or (2) the reward remains negative for at least t_{neg} consecutive steps.

In order to learn the optimal policy we exploit the Proximal Policy Optimization (PPO) algorithm, belonging to the family of reinforcement learning algorithms that use policy gradient methods. These methods optimize the expected cumulative reward by finding a good policy under the form of a neural network parameterized policy, thanks to variants of stochastic gradient ascent with respect to the policy parameters. A peculiarity of these approaches is their ability to work with large and continuous action spaces. We prefer policy gradient methods over value based methods, like Deep Q-Networks (DQN), since in our setting the number of actions corresponds to the overall number of features, a number which is typically large in the application domain of interest for our technique and the latter methods are not well-suited to deal with large action spaces.

The learning phase stops after a total number of T steps is performed. Once the policy has been learned, the optimal subspace has to be obtained from it. To this aim, a criterion to select informative subspaces is needed. The idea is to exploit the outcome of the learning phase in order to build a *ranking* among features, putting informative features first, from which a set of candidate subspaces can be obtained.

3.4 Subspace Rankings and Solution

The first ranking we consider is the *MDR Ranking*, for empirical Mean of Discounted Rewards, defined as:

$$\mathrm{MDR}(a) = \frac{1}{c(a)} \sum_{k,t \,:\, a_t^{(k)}=a} \gamma^t r_t^{(k)},$$

where $a_t^{(k)}$ ($r_t^{(k)}$, resp.) denotes the action performed (reward obtained, resp.) at the t-th time step of the k-th episode, and $c(a)$ is the action count of a, that is the number of times a is performed during learning. According to the *MDR Ranking*, the first feature f_1 is that with the highest MDR value, the second feature f_2 is that with the second highest MDR value, and so on.

The second ranking we consider, that we call *Policy Ranking*, can be derived from the learned policy π^*. According to this ranking, the i-th feature f_i ($1 \leq i \leq |F|$) is

$$f_i = \underset{f \in F \setminus \{f_1,\dots,f_{i-1}\}}{\arg\max}\ \pi^*(f \mid \{f_1, \dots, f_{i-1}\})$$

and, thus, the first feature f_1 is that with the highest probability $\pi^*(f \mid \emptyset)$.

Once a feature ranking $o = \langle f_1, \dots, f_d \rangle$ is fixed, a set $\mathcal{S}^o = \{S_1^o, \dots, S_d^o\}$ of d *candidate subspaces* is obtained, where $S_i^o = \{f_1, f_2, \dots, f_i\}$ ($1 \leq i \leq d$). Thus, according to each ranking o, we obtain the best explanation $\mathcal{E}^o$ as the subset of features S_i^o scoring the maximum gain

$$\mathcal{E}^o = \arg\max_{S_i^o} Gain(S_i^o)$$

The returned *optimum explanation* $\mathcal{E}^*$ is the explanation scoring the maximum $Gain$ among the considered rankings.

3.5 Subspace Refinement Strategy

In this section, we present an extended setting of our method, where the action space is enriched with two additional actions. The goal is to enable the method to perform *subspace refinement*.

Specifically, the action space is defined as follows: $A(S) = A_{\mathsf{add}}(S) \cup A_{\mathsf{rem}}(S) \cup \{\mathsf{stop}\}$, where $A_{\mathsf{add}}(S) = \{\mathsf{add}(f) \mid f \in F \setminus S\}$ and $A_{\mathsf{rem}}(S) = \{\mathsf{rem}(f) \mid f \in S\}$. The environment remains deterministic: if $a = \mathsf{add}(f)$ then the next state S' is $S' = S \cup \{f\}$, if $a = \mathsf{rem}(f)$ then $S' = S \setminus \{f\}$, and if $a = \mathsf{stop}$ the transition leads to a terminal state. In each state S, only actions in $A(S)$ are admissible, i.e., $\mathsf{add}(f)$ can be executed only if $f \notin S$, while $\mathsf{rem}(f)$ is available only if $f \in S$; stop is always available.

The availability of remove actions allows the agent to revise previously selected features and potentially escape locally suboptimal regions of the search space. Although the Subspace Margin can capture joint effects over the selected subset, RLAX explores the search space through single-feature modifications; strongly non-additive interactions may therefore be harder to discover when informative combinations require traversing lower-reward

intermediate subspaces. RLAX is thus better suited to settings where such combinations are reachable through locally improving or only moderately deteriorating transitions, while remove actions may partially mitigate this limitation.

Since the method is now able to refine the solution by removing features, in order to prevent the algorithm from entering a loop, we introduce a budget b, set to t_{max}, on the number of add operations. The reason why no fixed budget is imposed on the remove operations is that they do not create potential issues, as they are inherently upper-bounded by the number of add operations. Moreover, we want to allow the method to perform refinement freely, independently of the budget.

We introduce also the stop action to allow the method to avoid exploring suboptimal regions solely for the purpose of exhausting the budget and thus unnecessarily wasting time, giving it the ability to autonomously determine when to terminate.

Thus, episodes may terminate either by selecting stop, or by consuming the budget, or by the same negative-reward patience criterion above discussed (i.e., t_{neg} consecutive negative rewards). As for the reward, we set $R = 0$ for the stop action.

The refinement strategy induces a suitable MDR Ranking and Policy Ranking, that can be used as additional candidate subspaces for the computation of the optimal explanation $\mathcal{E}^*$.

4 Experimental Results

Experiments are organized as follows: Sect. 4.1 describes the setting used for the experiments. Section 4.2 analyzes RLAX behavior, including the *Gain* measure and the MDR ranking, statistical significance against random feature selection, and the impact of the Subspace refinement action space on training dynamics (episode length and budget-matched comparison). Section 4.3 compares RLAX with competitors. Section 4.4 investigates computational cost and scalability, while Sect. 4.5 evaluates RLAX on large-scale real-world datasets.

4.1 Experimental Settings

In the experiments, we considered a number of datasets from the ODDS Repository[1], namely *arrhythmia* (274 features and 406 instances), *cardio* (21 features and 1675 instances), *ionosphere* (33 features and 245 instances), *letter* (32 features and 1520 instances), *mnist* (100 features and 6923 instances), *optdigits* (64 features and 5086 instances), *pendigits* (16 features and 6734 instances), *satellite* (36 features and 4419 instances), and *satimage-2* (36 features and 5752 instances).

For each dataset, we generated n_v variants by randomly selecting n_e outlier instances as E_{pos} and n_e inlier instances as E_{neg}. All the remaining inlier instances form the reference set X, which is used to estimate the kNN-based anomaly scores. Thus, E_{neg} is disjoint from X. For evaluation, AUC is computed by considering the anomalous examples in E_{pos} as the positive class and the normal instances in X as the negative class. Therefore, the experimental protocol does not rely on a separate conventional train/test split.

[1] https://paperswithcode.com/dataset/odds

For each dataset variant, we performed n_r independent runs. For the best-$Gain$ analysis, we selected the run achieving the highest $Gain$ and retained its corresponding AUC and subspace dimension. Analogously, for the best-AUC analysis, we selected the run achieving the highest AUC and retained its corresponding $Gain$ and dimension. The final values reported for each dataset were then obtained by averaging these selected results over the n_v dataset variants. For stochastic methods, the same number of independent runs and the same best-run selection rule were applied. We used $n_e = 20$, $n_v = 10$, and $n_r = 10$.

As for the hyper-parameters of our method, t_{max} has been set to the maximum between $\sqrt{d}$, with $d = |F|$, and 30, while t_{neg} is a hyper-parameter that can be conveniently set to a small integer, we left it to 3. RLAX uses PPO from Stable-Baselines3 2.1.0 (Raffin et al., 2021) with a custom `ActorCriticPolicy` that masks inadmissible actions while retaining the default architecture: two 64-unit Tanh layers for both policy and value networks. Default PPO settings are used (learning rate $3 \cdot 10^{-4}$, $n_{\text{steps}} = 2048$, batch size 64, 10 epochs, $\lambda_{\text{GAE}} = 0.95$, clip range 0.2, entropy coefficient 0, value coefficient 0.5, and max gradient norm 0.5), except for $\gamma = 0.95$.

Throughout all the experiments reported in the following we use as distance measure the Euclidean distance, and as measure sep of separation to evaluate and compare results the AUC (Area Under the ROC Curve).

4.2 Analysis of the Method

In this section we analyze different aspects pertaining to the behavior of the method. Specifically, first we discuss the peculiarities of $Gain$ and MDR measures, then assess significance of results obtained by means of the proposed approach and, finally, investigate the behavior of the strategies in terms of average length of exploration episodes.

4.2.1 Performance Measures

Figure 1 compares the AUC and $Gain$ values obtained along the feature rankings induced by RLAX on *arrhythmia*, *mnist*, *letter*, and *optdigits*; the remaining datasets exhibit analogous trends. The figure highlights the solutions maximizing AUC and $Gain$, together with the AUC attained by the best-$Gain$ solution.

Although maximizing $Gain$ cannot yield an AUC higher than the AUC-optimal solution, it generally identifies substantially more compact subspaces while preserving comparable discriminative performance. In particular, the maximum AUC may require more than 30% of the available features, whereas the best-$Gain$ solution often involves only a small fraction of them. This confirms the ability of $Gain$ to balance separation quality and explanation compactness.

We further analyzed the sensitivity to the dimensionality penalty α. Increasing α progressively favors smaller explanations at the cost of a moderate decrease in AUC, while the default value $\alpha = 0.5$ provides an intermediate trade-off. Complete results are reported in Supplementary Table S3.

An additional analysis of the empirical Mean Discounted Reward (MDR) distributions is reported in the Supplementary Material.

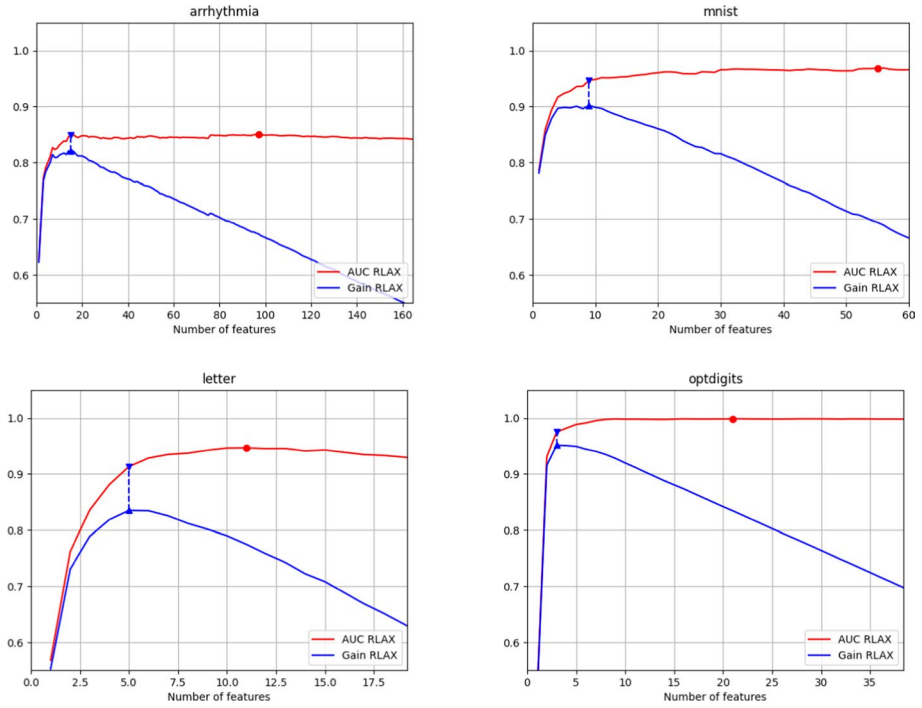


Fig. 1 Comparison between AUC and $Gain$

4.2.2 Significance of the Results

To assess the significance of the results obtained by RLAX, we compared it with *Random*, which selects feature subsets of a given size uniformly at random. Random results are averaged over the runs to estimate the expected quality of random feature selection.

Figure 2 compares the AUC of RLAX and Random on *arrhythmia*, *mnist*, *letter*, and *optdigits*. For RLAX, the curves are obtained by averaging across dataset variants the best AUC achieved at each subspace dimension.

Statistical significance was assessed through a one-sided paired Student's t-test. For each dataset variant, the best AUC across independent runs was selected at each dimension and the resulting curve was summarized by its mean AUC, yielding one paired observation per variant. The alternative hypothesis was that RLAX achieves higher mean AUC than Random. The resulting p-values were adjusted across the four datasets using Holm's correction.

The Holm-adjusted p-values are reported directly in Fig. 2 and confirm that RLAX significantly outperforms Random on all four datasets.

4.2.3 Behavior of the Strategies

Here we provide insights on the differences between the behavior of the two introduced strategies of RLAX. As for the episode length during training of the two strategies, namely the basic strategy and the subspace refinement strategy, it is highlighted in Fig. 3, where

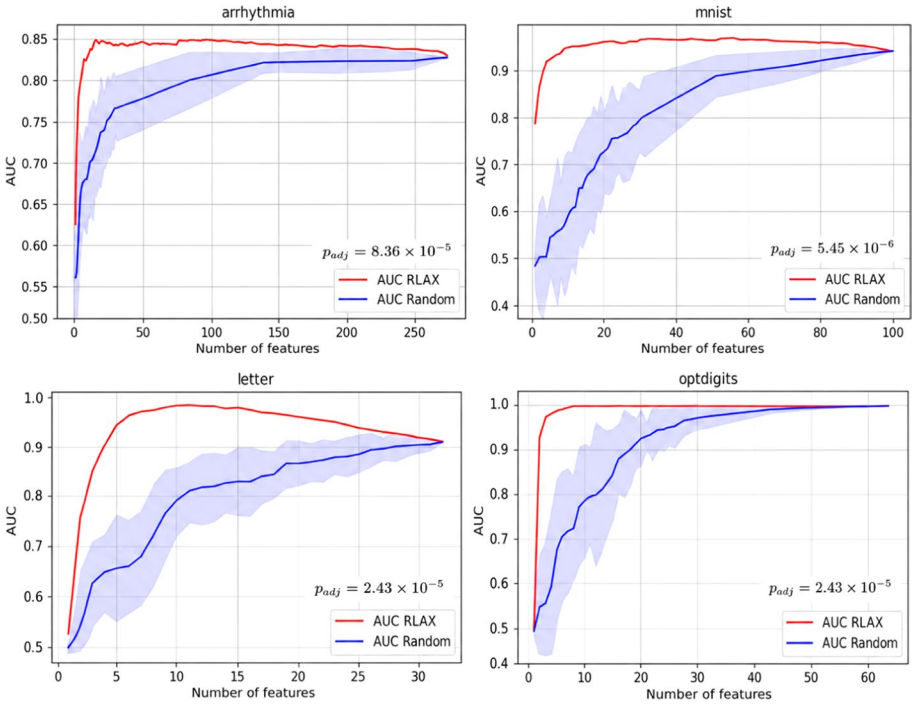


Fig. 2 Comparison between the AUC obtained by RLAX and Random feature selection

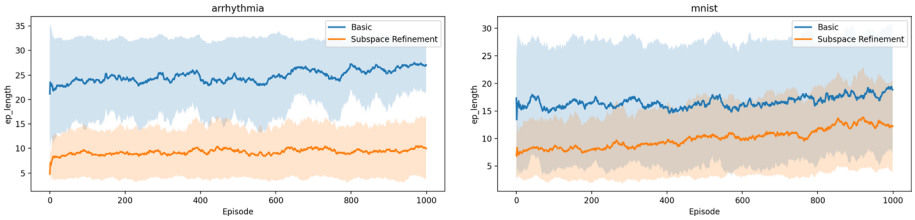


Fig. 3 Study of the episode length of *RLAX*

the first 1000 episodes are considered. On the representative datasets *arrhythmia* and *mnist*, the refinement setting consistently yields substantially shorter episodes, suggesting that the agent reaches a termination decision with fewer interactions and thus reduces the computational effort per episode. This behavior is consistent with the availability of stop and remove, which enable early termination and limited backtracking, in contrast to the basic setting where the policy can only expand the current subspace.

Figure 4 reports, for the two representative datasets *Ionosphere* and *mnist*, the AUC obtained from the features selected by the two strategies when trained using the same number of episodes, namely the number reached by the basic strategy with the default total number of time steps T. Note that, under this setting, the refinement strategy saves on average about 30% of the execution time. The plots show that, despite the reduced computational

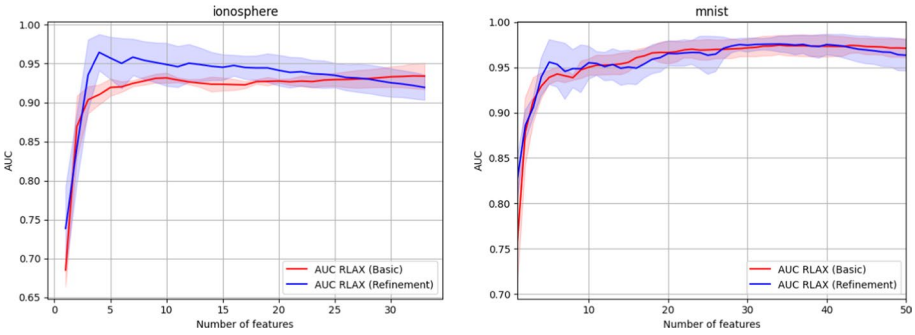


Fig. 4 Comparison between Basic strategy and Subspace Refinement strategy

effort, the subspace refinement strategy achieves improved AUC performance for the same feature subset size. Notably, this improvement is observed even for the top-ranked features.

4.3 Comparison with Competitors

We compared RLAX with V-GAN (Cribeiro-Ramallo et al., 2025), SOD (Kriegel et al., 2009), HiCS (Keller et al., 2012), Genetic (Fassetti & Angiulli, 2010), and Random. For methods returning multiple subspaces, the reported value for each run is the average over the corresponding subspaces.

Table 1 reports AUC, *Gain*, and subspace dimension for the solutions selected according to the best *Gain*. The best *Gain* for each dataset is highlighted in bold, while selected AUC and dimensionality values are underlined. RLAX achieves the highest *Gain* on all datasets, indicating a consistently favorable trade-off between discriminative performance and explanation compactness. Genetic occasionally attains slightly higher AUC values, but typically requires substantially larger subspaces, whereas HiCS, and in one case Random, may return smaller subspaces at the cost of lower AUC.

The corresponding comparison based on the best AUC is reported in the Supplementary Material.

We also assessed the stability of the feature subsets selected by RLAX using the Nogueira stability measure (Nogueira et al., 2018). As shown in Table 2, the selected subsets exhibit good to excellent stability across all datasets.

4.4 Computational Cost and Scalability

To investigate the computational cost of RLAX, we performed an additional controlled experiment on synthetic data by varying both the number of instances, $n \in \{100, 1000, 2000\}$, and the number of input features, $d \in \{16, 64, 256\}$. The same datasets were used for RLAX and all competing methods.

End-to-end wall-clock execution times and the corresponding explanation-quality results are reported in the Supplementary Material.

Figure 5 reports the relative runtime growth as the number of instances increases for $d = 16$, 64, and 256. Although RLAX has a higher absolute computational cost due to PPO training, its runtime grows moderately: from $n = 100$ to 2000, it increases by approximately

Table 1 Comparison on best *Gain*

Method	arrhythmia			cardio			ionosphere		
	AUC	Gain	Dim	AUC	Gain	Dim	AUC	Gain	Dim
RLAX	0.872 (.024)	0.855 (.010)	9.0 (7.9)	0.961 (.047)	0.885 (.044)	3.2 (2.2)	0.970 (.032)	0.919 (.028)	3.4 (0.9)
Random	0.786 (.036)	0.754 (.032)	16.6 (7.09)	0.846 (.025)	0.703 (.045)	5.0 (1.58)	0.889 (.025)	0.820 (.017)	3.6 (1.52)
Genetic	0.884(.009)	0.653 (.028)	127.0 (13.94)	0.970(.007)	0.670 (.017)	12.6 (0.80)	0.949 (.006)	0.606 (.014)	22.6 (0.80)
HiCS	0.740(.028)	0.723(.028)	9.8(0.17)	0.855(.007)	0.718(.006)	5.8(0.08)	0.861(.013)	0.794(.013)	4.4(0.05)
SOD	0.743 (.033)	0.360(.012)	210.2(1.25)	0.817 (.046)	0.460(.004)	15.0(0.00)	0.939 (.028)	0.667(.037)	18.0(0.00)
V-GAN	0.708 (.031)	0.478 (.030)	126.5 (4.22)	0.385 (.088)	0.128 (.052)	10.8 (1.55)	0.882 (.016)	0.628 (.024)	16.8 (1.87)
Method	letter			mnist			optdigits		
	AUC	Gain	Dim	AUC	Gain	Dim	AUC	Gain	Dim
RLAX	0.942 (.022)	0.848 (.021)	6.0 (0.96)	0.974 (.012)	0.948 (.008)	5.2 (0.98)	0.996 (.004)	0.973 (.003)	2.9 (0.70)
Random	0.812 (.062)	0.665 (.029)	8.4 (3.13)	0.820 (.066)	0.714 (.030)	20.2 (7.60)	0.955 (.033)	0.819 (.016)	16.4 (4.16)
Genetic	0.948(.016)	0.745(.021)	13.0(0.89)	0.968(.012)	0.753(.006)	43.0(1.67)	1.000(.00007)	0.772(.010)	29.2(1.33)
HiCS	0.704(.035)	0.634(.035)	4.5(0.05)	0.638(.014)	0.596(.015)	8.4(0.17)	0.789(.007)	0.741(.006)	6.1(0.00)
SOD	0.916 (.016)	0.663(.021)	16.2(0.47)	0.864 (.067)	0.474(.037)	78.0(0.00)	0.749 (.051)	0.449(.046)	38.4(0.48)
V-GAN	0.485 (.050)	0.230 (.049)	16.3 (1.89)	0.569 (.058)	0.322 (.059)	49.4 (2.37)	0.495 (.010)	0.241 (.024)	32.6 (2.59)
Method	pendigits			satellite			satimage-2		
	AUC	Gain	Dim	AUC	Gain	Dim	AUC	Gain	Dim
RLAX	0.997 (.004)	0.928 (.014)	2.2 (0.41)	0.935 (.056)	0.891 (.058)	3.2 (1.06)	0.999 (.012)	0.971 (.017)	2.0 (0.92)
Random	0.942 (.023)	0.804 (.018)	3.4 (1.14)	0.864 (.030)	0.822 (.040)	2.0 (1.22)	0.982 (.013)	0.943 (.009)	1.8 (0.45)
Genetic	0.999(.00013)	0.593 (.034)	13.0 (1.10)	0.909 (.016)	0.576 (.014)	24.0 (1.41)	1.000(.00074)	0.677(.005)	23.2(0.40)
HiCS	0.932(.010)	0.778(.009)	4.9(0.00)	0.865(.040)	0.800(.040)	4.7(0.04)	0.978(.011)	0.912(.010)	4.7(0.04)
SOD	0.903 (.027)	0.653(.032)	8.0(0.10)	0.844 (.053)	0.633(.012)	15.2(0.15)	0.813 (.080)	0.604(.013)	15.0(0.00)
V-GAN	0.691(.001)	0.435(.031)	8.2(1.03)	0.822(.009)	0.564(.029)	18.6(2.17)	0.955(.0003)	0.697(.030)	18.6(2.17)

Values are reported as mean (standard deviation)

Table 2 Nogueira stability of RLAX feature subsets

Dataset	Nogueira index	Level
arrhythmia	0.555 ± 0.150	Good
cardio	0.958 ± 0.064	Excellent
ionosphere	0.898 ± 0.140	Excellent
letter	0.901 ± 0.123	Excellent
mnist	0.893 ± 0.065	Excellent
optdigits	0.837 ± 0.104	Excellent
pendigits	0.766 ± 0.254	Excellent
satellite	0.532 ± 0.164	Good
satimage-2	0.591 ± 0.266	Good

We adopt the scale Good: $0.40 \leq \Phi < 0.75$; Excellent: $\Phi \geq 0.75$ proposed in the original paper (Nogueira et al., 2018)

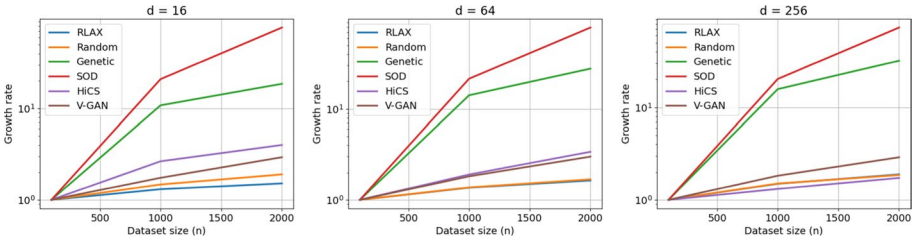


Fig. 5 Runtime growth as the dataset size increases for $d = 16$, $d = 64$, and $d = 256$

1.5×, 1.6×, and 1.9×, respectively. This growth is substantially lower than that of Genetic and SOD and comparable to the other competitors.

As reported in the Supplementary Material, the higher computational cost is accompanied by better explanation quality: RLAX achieves the highest average AUC (0.979) and Jaccard agreement (0.669), while retaining compact explanations with an average dimension of 3.48.

4.5 Experiment on Real-World Datasets

To evaluate the robustness and generalization of our approach in realistic scenarios, we consider two additional real-world datasets, *Swarm Behaviour* and *Internet Advertisements*, which are substantially larger, high-dimensional and more heterogeneous in their feature representations.

We start by providing details on the two data domains:

Internet Advertisements: The Internet Advertisements dataset contains instances describing web-page images and related textual/URL-derived attributes. The task is to discriminate advertisements (`ad`) from non-advertisements (`nonad`); the dataset has 1, 558 features and includes missing values. We treat one class as the anomalous group (positive samples) and the other as the inlier population (negative samples).

Swarm Behaviour: The Swarm Behaviour dataset (UCI) models collective motion patterns through attributes repeated for a fixed number of agents (boids) and comprises 2, 400 features, with binary labels indicating whether the swarm exhibits specific emergent properties (e.g., flocking/alignment) versus the corresponding negative condition. We cast the problem into our anomaly-explanation setting by selecting one label as the anomalous group

Table 3 Comparison on real-world datasets

Method	Swarm_Behaviour			InternetAds		
	AUC	Gain	Dim	AUC	Gain	Dim
RLAX	0.915	0.914	6.00	0.873	0.870	9.5
Genetic	0.884	0.650	1124.0	0.777	0.554	694.5
HiCS	0.700	0.700	2.10	0.536	0.536	2.05
SOD	0.876	0.120	2353.50	0.858	0.364	1539.00
V-GAN	0.626	0.624	6.30	0.827	0.822	14.32

and the other as inliers, consistently with the protocol used in the rest of the experimental section.

We follow the same experimental protocol adopted in the previous sections, generating multiple dataset variants by sampling E_{pos} and E_{neg}. Performance is evaluated using AUC and *Gain*.

Table 3 reports the results on the two large-scale real-world datasets. RLAX achieves a favorable balance between separability and compactness: HiCS (Keller et al., 2012) returns smaller subspaces but lower AUC, whereas SOD (Kriegel et al., 2009) and Genetic (Fassetti & Angiulli, 2010) achieve stronger separability with substantially larger explanations. Overall, *Gain* provides an explicit balance between discriminative performance and explanation compactness.

4.6 Qualitative Analysis of Extracted Explanations

To address the interpretability and practical usefulness of the explanations produced by our method, we complement the quantitative evaluation with a qualitative analysis of the feature subsets selected by RLAX. We focus on the *cardio* dataset from the ODDS repository, as each of its attributes carry a clear clinical semantics, making it particularly suitable for an interpretability-oriented case study.

For a representative anomaly in this dataset, RLAX selects the following top features as most relevant to the anomaly decision: *DP* (number of prolonged decelerations per second), *Variance* (histogram variance of the fetal heart rate), *ALTV* (percentage of time with abnormal long-term variability), and *FM* (number of fetal movements per second). We discuss each of these below, relating them to established findings in the obstetric and cardiotocography (CTG) literature.

Prolonged decelerations (DP) are well documented in the clinical literature as a strong indicator of fetal pathological status. They are defined as sustained drops in fetal heart rate (FHR) exceeding 15 bpm for more than 3 minutes; when they persist beyond 5 minutes with FHR below 80 bpm and reduced variability, they signal acute fetal hypoxia/acidosis requiring urgent intervention. Together with late decelerations, prolonged decelerations are the only deceleration type considered inherently indicative of serious fetal compromise, in contrast to early decelerations (benign) and variable decelerations (generally innocuous) (Cahill et al., 2018; Fogelberg et al., 2025). The selection of DP among the top explanatory features is therefore consistent with its recognized clinical role as a marker of fetal distress.

Although *histogram variance (Variance)* is a feature computationally derived from the FHR histogram within the SisPorto system and does not correspond to a single standardized clinical finding, it is conceptually related to the clinical notion of FHR oscillation amplitude

(bandwidth), a parameter historically included in CTG scoring criteria (e.g., the Kubli and Fischer scores) and still clinically relevant today. Both reduced variability (silent pattern, associated with hypoxia/acidosis) and excessive variability (saltatory pattern, also associated with acute hypoxia) are regarded as non-reassuring signs. The selection of this feature by RLAX is thus plausibly explained by its ability to capture deviations in FHR dispersion in both directions, consistent with the underlying pathophysiology.

Abnormal long-term variability (ALTV) is one of the four fundamental characteristics used to interpret a CTG trace, together with baseline, accelerations, and decelerations. It reflects slower oscillations of the fetal heart rate (approximately 3–10 cycles/minute, amplitude 10–25 bpm) and is an expression of the integrity of the fetal autonomic nervous system. ALTV specifically quantifies the percentage of recording time during which this variability is abnormal, making it a direct quantification of a standard clinical finding (Tarvonen et al., 2022; Williams & Galerneau, 2003).

Fetal movements (FM), whether perceived by the mother or instrumentally recorded (as in the non-stress test), are a well-established indicator of fetal well-being. A reduction in fetal movements is a recognized clinical warning sign, historically used as a screening test ("kick counts") (Bradford et al., 2024; Hantoushzadeh et al., 2024).

Overall, all top four features selected by RLAX correspond to established, clinically meaningful indicators of fetal distress, spanning both direct CTG findings (DP, ALTV, FM) and a derived statistic with a plausible clinical interpretation (Variance). This alignment between the automatically selected explanatory features and domain knowledge supports the interpretability and practical usefulness of the explanations produced by our method.

5 Conclusions

We addressed the Anomaly Explanation problem as a feature selection task aimed at identifying compact and interpretable subspaces that discriminate anomalous examples from normal data. To this end, we introduced RLAX, a reinforcement learning framework that formulates explanation search as a Markov Decision Process and exploits Subspace Margin as a dense training signal.

This work extends our previous version (Amirato et al., 2025) through both methodological and empirical advances. In particular, the Subspace Refinement Strategy enriches the action space with explicit *remove* and *stop* actions, enabling more flexible exploration beyond purely additive feature selection. Extensive experiments on benchmark and real-world datasets show that RLAX consistently identifies compact explanations while preserving strong discriminative performance.

Overall, the results support reinforcement learning as an effective strategy for exploring the feature-subset space underlying anomaly explanation. Future work will investigate domain-informed reward shaping, alternative separation measures, and broader settings involving heterogeneous features and weaker supervision.

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Data Availability All datasets used in this study are publicly available. Standard benchmarks were obtained from the ODDS Repository (https://paperswithcode.com/dataset/odds). The Swarm Behaviour and Internet Advertisements datasets are available at the UCI Machine Learning Repository (https://archive.ics.uci.edu).

Code Availability The source code of RLAX is publicly available at https://github.com/AIDALab-DIMES/RLAX.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

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